



XMBA 201A
Economic Analysis for Business Decisions

A Classification of Economic Costs and Introduction to Price Setting

Lecture 3

Professor Steve Tadelis

1

Recap: Economic Costs vs. Accounting Costs

- There are two things to do with expenses:
 - Keep track of them (EXPENSES: RECORD KEEPING).
 - Use them for decisions (COSTS: DECISION MAKING).
- Accounting rules are designed to help firms & investors keep track of **expenses**: where did the money go? Where did it come from?
- A **first key difference** between economists and accountants is that accountants are less able to recognize opportunity costs.
- A **second key difference** between economists and accountants is that accountants are not allowed to ignore sunk costs.
- Separating accounting objectives from decision objectives is critical!!

2

Any dark chocolate lovers?

Imagine that I have
10 pieces of really
good dark chocolate!



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Page 3

3

Bidding for chocolate?

- **Question:** How much are you willing to pay for it **here and now**?
- If you tell me, I have a bunch of “willingness to pay” (WTP) data points
- The bidding process
 1. You’re bidding for *one piece of wrapped chocolate*
 2. The *top-10 bidders win a piece of chocolate each*
 3. Each winner pays the 11th highest bid
(which is *the highest “losing” bid*)
 1. If there are ties at no. 10, then I’ll randomly select the marginal winners (and the highest losing price is one of the tied 10th highest price bids)
- Bidding is open!

SCAN ME



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Page 4

4

Articles

- “Chinese Solar Farms Are Crowding Out Much-Needed Crops”
- "He Earns \$33 an Hour as a Costco Cashier. Now He's a Millionaire"Download He Earns \$33 an Hour as a Costco Cashier. Now He's a Millionaire"
- “Amazon’s new warehouse policy will penalize sellers for storing products too long”
 - Why was there a need to create the new policy?
 - In what way does the new IPI score account for the costs that third-party sellers impose on Amazon’s warehouse?

5

The Shared Overhead Fallacy

- You offer High School tutoring in two locations:
 - In Berkeley, where you live, and get \$30 an hour
 - In Palo Alto, where there are rich kids, and you get \$40 an hour.
- Your alternative workplace is the Berkeley Library, where you get \$20 an hour, and work is abundant.
- There are 5 hours a day worth of tutoring in Berkeley, and 2 hours in Palo Alto. The costs of commuting are 3 hours on the road, and \$25 in tolls, gas and wear & tear.
- How is your business doing?

6

The Shared Overhead Fallacy

- Daily income (10 hours) = $5 \times 30 + 2 \times 40 - 25 = \205
- Your accountant understands opportunity costs and claims:
 - You are losing 3 hours of work a day on the road worth $3 \times 20 = \$60$ (lost library income), and \$25 in tolls, gas, wear and tear, making a total of \$85 in overhead for your business.
 - Your revenues are \$230, about 65% from Berkeley and 35% from Palo Alto. Thus, the overhead allocated to the Berkeley activities are $0.65 \times 85 = \$55.25$, and to Palo alto, $0.35 \times 85 = \$29.75$
 - Your **profit** from each activity is therefore:
 - Berkeley net income = $5 \times (30 - 20) - \$55.25 = \$ -5.25$
 - Palo Alto net income = $2 \times (40 - 20) - \$29.75 = \$ 10.25$

Potential
Library
income

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Page 7

7

The Shared Overhead Fallacy

- Your accountant recommends:
 - Keep Palo Alto, drop the Berkeley business and spend those 5 hours in the library!**
- Sounds like sound advice?
 - There is no reason that overhead should be allocated in proportion to generated revenues
 - In fact, overhead should, when possible, be allocated to the **activity that causes it!** (causal allocation)
 - Here all the commute is coming from Palo Alto!
 - Berkeley **net** income = $5 \times (30 - 20) = \$ 50$
 - Palo Alto **net** income = $2 \times (40 - 20) - \$85 = \$ -45 !$

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Page 8

8

Economic Costs vs. Accounting Costs: Recap

- There are two things to do with expenses:
 - Keep track of them (EXPENSES: RECORD KEEPING).
 - Use them for decisions (COSTS: DECISION MAKING).
- Accounting rules are designed to help firms & investors keep track of **expenses**: where did the money go? Where did it come from?
- Accountants are less willing to recognize uncertain costs (and often opportunity costs).
- Accountants are unwilling to ignore previous (sunk) expenditures. This is a **second key difference** between economists and accountants

Economic Costs: Take Aways

- Economic costs:

Costs = expenditures – sunk costs + opportunity costs
- Test for sunk costs:

If you can't affect an expenditure over the relevant decision-making horizon, it's a sunk expenditure.
- Overhead (fixed) cost allocation:

If you do any of it, it should be causal. If it can't be causally allocated, don't allocate it at all!

A Categorization of Costs

- There are two common categories of costs:
 - **Variable costs:** these are costs incurred per unit of production (raw material, direct labor, etc.)
 - **Overhead (or fixed) costs:** these are costs incurred by the activity, but that do not vary with production (machine leases, HR personnel, etc.)
- Overhead costs can be tricky:
 - they may be **sunk** (R&D, past rent, training,...)
 - they may be caused by **one activity** (production machine, energy,...)
 - they may be caused by several activities (R&D, support staff, rent,...)
 - they may be shared by the **whole company!** (C-suite execs, acting,...)

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Page 11

11

A Categorization of Costs

- **Total Cost:** the cost of producing some number of units of output: overhead and variable. e.g., to produce q tires it costs:

$$C(q) = F + V(q);$$

F = fixed costs; $V(q)$ = variable

- **Average Cost:** $AC(q) = C(q)/q$
- **Marginal Cost:** the cost of producing the next unit. The marginal cost of the n^{th} unit is,

$$\begin{aligned} MC(q) &= C(q) - C(q - 1) \\ &= [F + V(q)] - [F + V(q - 1)] \\ &= V(q) - V(q - 1) \end{aligned}$$

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Page 12

12

Articles

- “Thanks to Nvidia, AI Will Soon Take Your Order at Taco Bell and Pizza Hut”
 1. What are the issues that determine the costs and benefits from adopting AI for drive-thru orders?
 2. How would waiting for the technology to improve impact these costs and benefits?
 3. How does this adoption change fixed versus variable costs?

13

A Categorization of Costs: Deep Dive Example

- As argued earlier, it is good to know where costs come from, but ***what matters for decision making?***
- **Example:** Your division manager tells you that the market price of combs is \$1, and that the average cost of a comb produced is 90 cents, so the profits are \$1,000 a day for the 10,000 combs produced a day, a 10% return.
- Are you happy?

14

A Categorization of Costs: Deep Dive

- Upon further scrutiny you learn that
 1. Machine and space (overhead) costs are \$500
 2. Material costs are 30 cents a comb
 3. Labor costs are \$500 an hour for the first 8 hours and time-and-a-half afterward.
 4. Each hour of production generates 1000 combs
- Are you happy?

15

The Average Cost Fallacy

- The first 8000 combs cost

$$C(q) = F + V(q) = \text{\$500} + \text{\$2,400} + \text{\$4,000} = \text{\$6,900}$$

fixed
material
labor
- The profit on these is \$1,100
- The next 2000 combs cost
 - $\text{\$600} + \text{\$1,500} = \text{\$2,100}$

material
labor
- Should you be happy? **NO!**
 - There is a profit of \$1,100 on the first \$8000 produced
 - But there is a loss of \$100 on the last 2000 produced!

16

The Average Cost Fallacy

- The problem is a subtle one:
- If average costs are below the price per unit then you are making a **positive profit**.
 - Average cost per unit is the average cost (DUH...), and revenue per unit is the price
- You are making a **per unit** profit **on average**, but not necessarily **on every unit sold**
- Each of the last 2,000 units cost \$1.05 to produce versus a price per unit of \$1 !

17

A Mathematical and Graphical Representation

- Recall the fundamentals:
 - Each hour of production generates 1000 combs
 - Machine and space: \$500
 - Material: \$0.30 per unit
 - Labor:
 - \$500/1000 = \$0.50 for the first 8000 units
 - \$750/1000 = \$0.75 for every unit thereafter

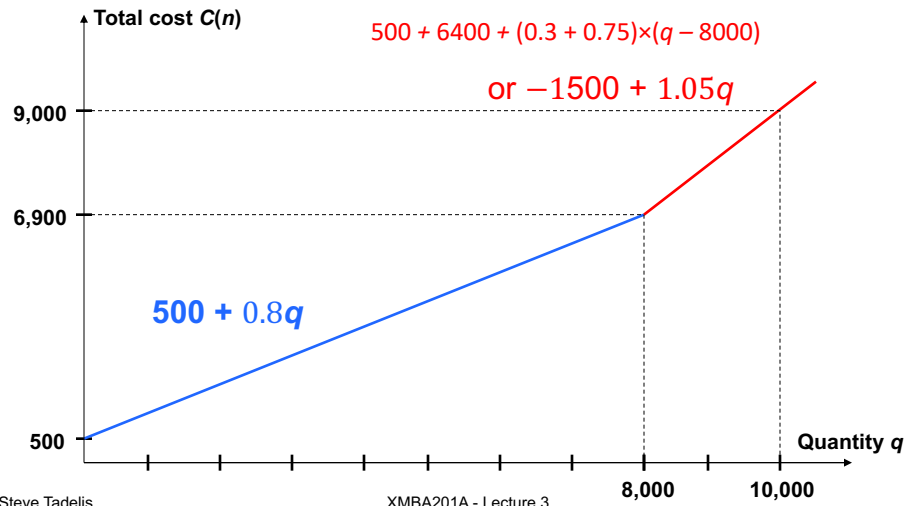
$$C(q) = \begin{cases} 500 + (0.3 + 0.5) \times q & \text{if } q \leq 8,000 \\ 500 + 6,400 + (0.3 + 0.75) \times (q - 8,000) & \text{if } q > 8,000 \end{cases}$$


 $= (0.3 + 0.5) \times 8,000$

18

A Mathematical and Graphical Representation

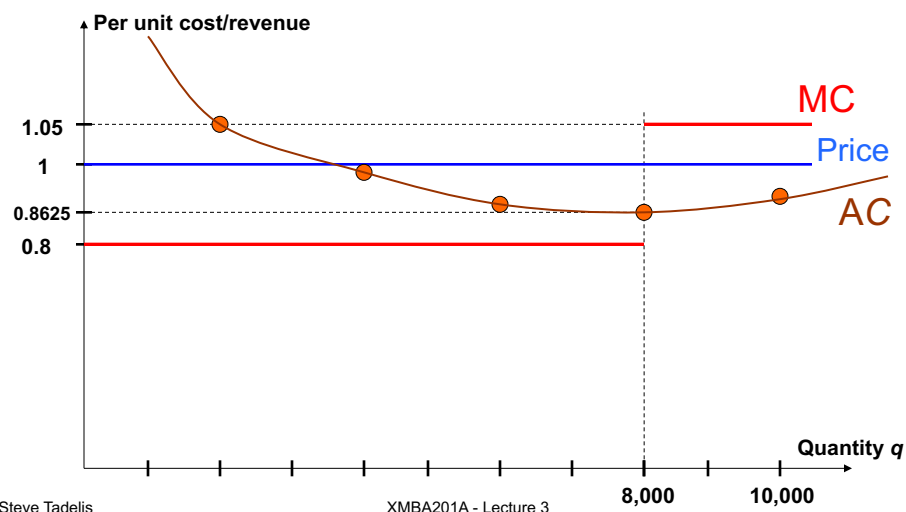
- We can graph $C(n)$:



19

A Mathematical and Graphical Representation

- We can graph $AC(n)$, $MC(n)$ and the price:



20

Optimal Behavior

- What should the firm do?
- Ignoring fixed costs (overhead), the firm is making \$0.2 on every comb up to 8,000 combs because for $q < 8,000$:

$$MC(q) = 0.8 < 1 = p$$

- Similarly, it is losing \$0.05 on every comb beyond the first 8,000 because for $q > 8,000$:

$$MC(q) = 1.05 > 1 = p$$

- What seems to be the general lesson?

21

Optimal Behavior

- **What to produce**: every unit for which the income on that unit (in this case the price) is greater than the cost
- **What not to produce**: any unit for which the income on that unit (in this case the price) is less than the cost
- This simple rule includes the intuition for monopoly pricing going beyond a fixed price and a linear cost function.
- **NOTE**: $AC(q)$ is not playing a role for how much to produce, ***only for if to produce at all!***

22

Summary

- Average costs are important for **operate/shut-down** decisions, but **not for optimal quantity** decision!
- Marginal costs are important for **optimal quantity decisions!**
- Costs needs to be allocated to the decision that they influence in ways that reflect how the decision affects the cost!

23

Introduction to Pricing

24

In General: What is a Firm's Problem?

- **Objective:** Maximize Profits
- **Alternatives:**
 - **Choose Quantity:** A firm can choose to dump a quantity and see at which price it can sell it all.
 - **Choose Price:** The firm may be able to choose a price and sell quantities up to the implied demand
- **The route from Alternative to the Objective:**
 - The price-quantity pair determines revenues ($p \times q$)
 - The quantity determines costs ($C(q)$)
 - Profit = revenue – costs = $p \times q - C(q)$!

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Page 25

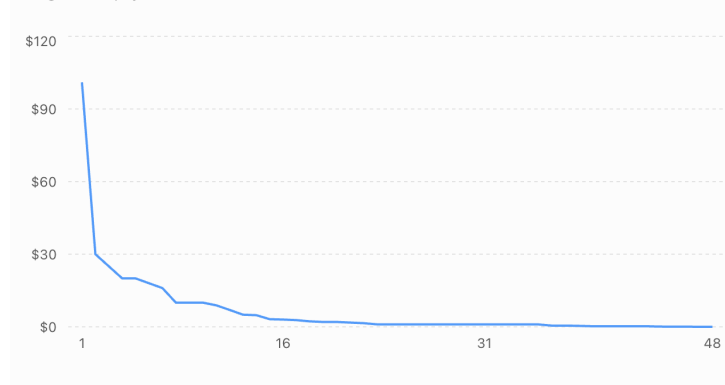
25

The Origins of Demand: Chocolate Example

- You each submitted your “willingness to pay”. What does that give me?

Demand curve from willingness-to-pay data

Each observation represents one potential buyer of one unit. Price is the marginal buyer's willingness to pay.



	Name	WTP	Revenue
1	Soheil	101.00	30
2	Bindu Sheth	30.00	50.02
3	Mike D	25.01	60.03
4	Sayf	20.01	80
5	Christian	20.00	90
6	Steve Weisbrod	18.00	96
7	Yui	16.00	70
8	Sushruth	10.00	80
9	BK Abebe	10.00	90
10	Sushruth	10.00	88.9
11	Jayne Laiprasert	8.89	77
12	baran	7.00	60.12
13	Ravi	5.01	

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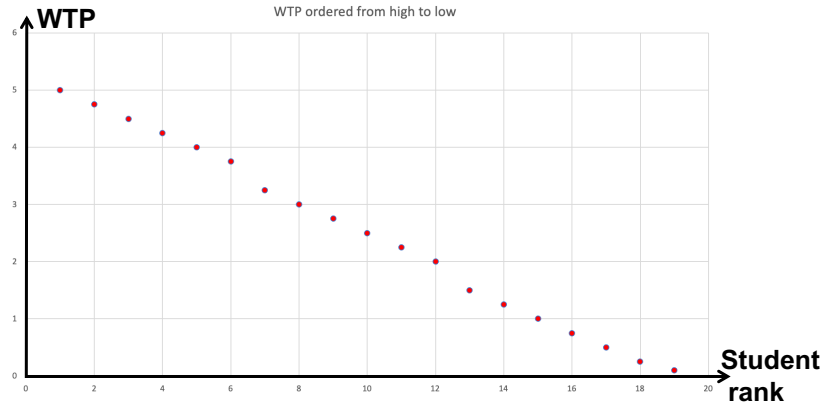
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Page 26

26

“Marginal” Willingness to Pay Plot

- **Fake Data:** Imagine that only 19 people like chocolate and their willingness to pay is (highest to lowest):



- **How do I maximize profits?**

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Page 27

27

Charging each individual separately

- If I could, I would charge \$5 from student 1, \$4.75 from 2, and so on to reap \$47.35!
- Is that easy to do?
 - Student 1 will not be willing to buy and will wait for me to sell to students 2 – 19, the last paying \$0.1, to which student 1 may offer to buy for, say, \$3... (**arbitrage!**)
- **Implication:** must set one price for all!!
“simple (monopoly) pricing”
- **How do I choose the price?**

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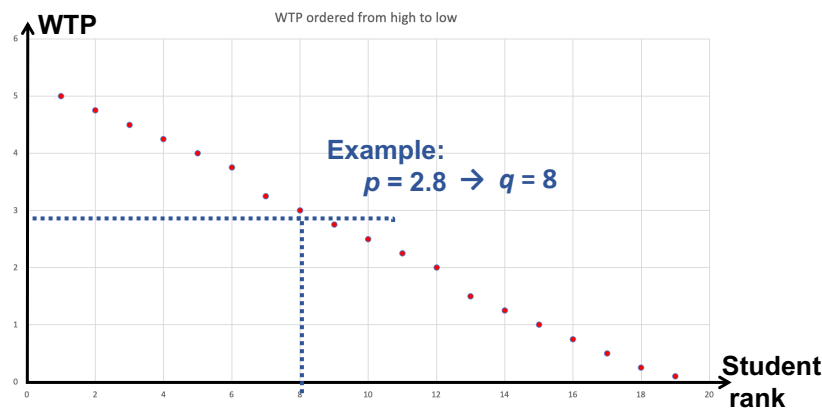
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Page 28

28

Demand Functions: from Many to One

- Marginal WTP is the demand function! For any price, how much will I sell?



5
4.75
4.5
4.25
4
3.75
3.25
3
2.75
2.5
2.25
2
1.5
1.25
1
0.75
0.5
0.25
0.1
0

- Unique to economics: $p(q)$ instead of $q(p)$ [will be very convenient!]

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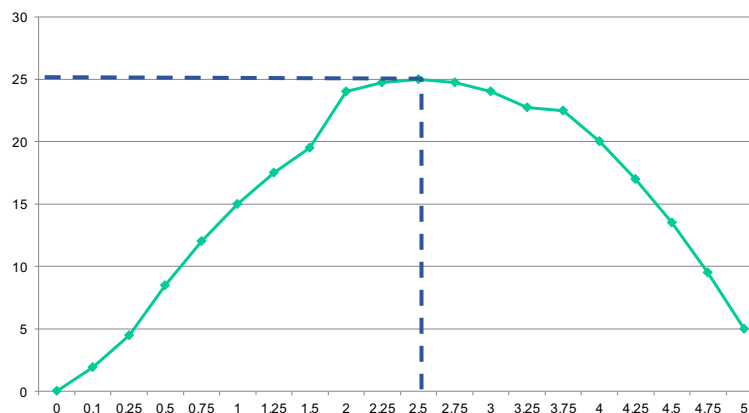
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Page 29

29

Revenue Generation

- We can plot the revenue as a function of price:



5
4.75
4.5
4.25
4
3.75
3.25
3
2.75
2.5
2.25
2
1.5
1.25
1
0.75
0.5
0.25
0.1
0

- **Question:** What *price* should we set? **$P = 2.5$**

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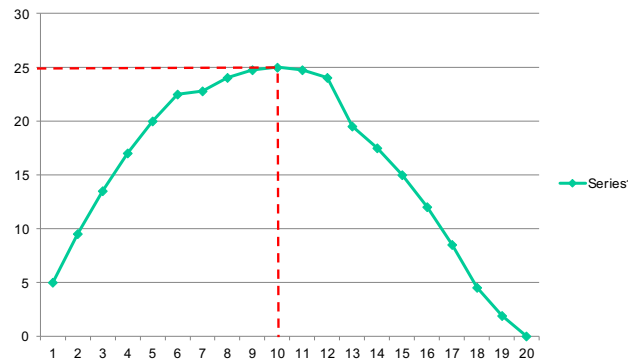
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Page 30

30

Revenue Generation: Another Look

- Revenue as a function of quantity:



Quantity	WTP/p	Revenue
1.00	5	5
2.00	4.75	9.5
3.00	4.5	13.5
4.00	4.25	17
5.00	4	20
6.00	3.75	22.5
7.00	3.25	22.75
8.00	3	24
9.00	2.75	24.75
10.00	2.5	25
11.00	2.25	24.75
12.00	2	24
13.00	1.5	19.5
14.00	1.25	17.5
15.00	1	15
16.00	0.75	12
17.00	0.5	8.5
18.00	0.25	4.5
19.00	0.1	1.9
20.00	0	0

- Question:** how much we should sell? $q = 10$
- Quantity/Price are determined together – but looking at q is more revealing...

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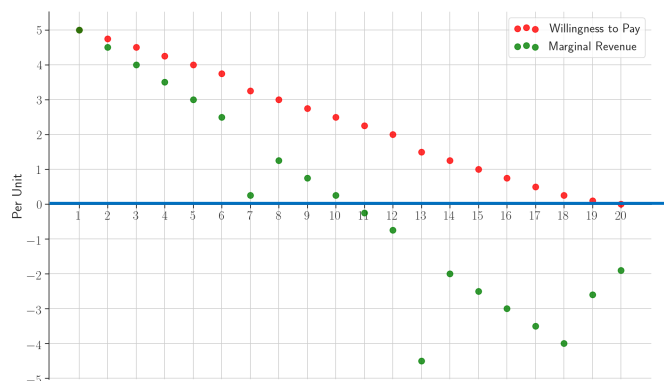
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Page 31

31

Optimal Production/Pricing

- Demand** and the **Marginal Revenue** Curves:



Quantity	WTP/p	Revenue	MR
1.00	5	5	5
2.00	4.75	9.5	4.5
3.00	4.5	13.5	4
4.00	4.25	17	3.5
5.00	4	20	3
6.00	3.75	22.5	2.5
7.00	3.25	22.75	0.25
8.00	3	24	1.25
9.00	2.75	24.75	0.75
10.00	2.5	25	0.25
11.00	2.25	24.75	-0.25
12.00	2	24	-0.75
13.00	1.5	19.5	-1.5
14.00	1.25	17.5	-2
15.00	1	15	-2.5
16.00	0.75	12	-3
17.00	0.5	8.5	-3.5
18.00	0.25	4.5	-4
19.00	0.1	1.9	-4.5
20.00	0	0	-5

- How much to produce depends on **MARGINAL COSTS!**
- Easy case: $MC = 0 \rightarrow q = 10$ [Maximize revenue: $R(q)$]
- Does this remind you of anything? **Combs!**

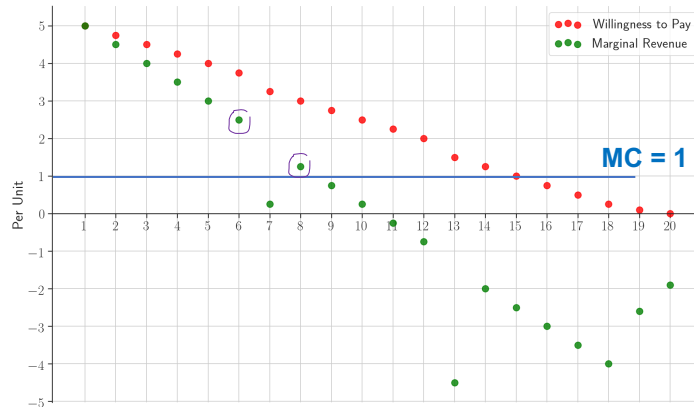
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Page 32

32

Optimal Production/Pricing: Harder case: $MC = 1$



Quantity	WTP/p	Revenue	MR
1.00	5	5	5
2.00	4.75	9.5	4.5
3.00	4.5	13.5	4
4.00	4.25	17	3.5
5.00	4	20	3
6.00	3.75	22.5	2.5
7.00	3.25	22.75	0.25
8.00	3	24	1.25
9.00	2.75	24.75	0.75
10.00	2.5	25	0.25
11.00	2.25	24.75	-0.25
12.00	2	24	-0.75
13.00	1.5	19.5	-4.5
14.00	1.25	17.5	-2
15.00	1	15	-2.5
16.00	0.75	12	-3
17.00	0.5	8.5	-3.5
18.00	0.25	4.5	-4
19.00	0.1	1.9	-2.6
20.00	0	0	-1.9

- Need to check two points: $q = 6$ and $q = 8$
 - $q = 6 \Rightarrow p = 3.75, \Pi = 6 \times (3.75 - 1) = 16.5$
 - $q = 8 \Rightarrow p = 3, \Pi = 8 \times (3 - 1) = 16$

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Page 33

33

Summary

- **How much to sell (price):** so that every unit for which the income on that unit (marginal revenue) is greater than the marginal cost
- This simple rule is the backbone for monopoly pricing.
- Therefore, MC and demand (MR) determine the optimal price/quantity
- $AC(q)$ is not playing a role for how to price or how much to produce, **only matters for if to produce at all!**

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Page 34

34